dynatrace

5 challenges to achieving observability at scale

AI moves the goal posts for every IT role



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INTRODUCTION

Every business is a digital business

Digital platforms are the backbone of modern business. In 2026 and beyond, organizations will need more than just visibility into their systems. They will need actionable insights that reveal how system performance impacts business outcomes in real time. Performance metrics, user experience, and operational health must directly connect to outcomes such as revenue, resilience, and innovation. But the real challenge isn't gathering data; rather, it is transforming a flood of signals into clear, actionable decisions that drive measurable business impact.

Complexity has evolved

Complexity didn't stop with cloud migration. Organizations are still navigating hybrid environments, ephemeral instances, and mountains of telemetry, but now a new layer has emerged: the AI boom. Generative, predictive, causal and agentic AI systems are everywhere, acting autonomously and influencing critical business outcomes. More data does not equal better decisions; in fact, it amplifies noise. Context is now essential.

AI-powered observability is the only way to keep pace with this scale, but when AI makes the wrong call, the consequences ripple across the business. Observability must evolve beyond monitoring and dashboards into a strategic asset and intelligent control plane: one that governs AI decisions, enforces transparency, and ensures automation acts in alignment with business goals.

The emergence of autonomous operations

Modern observability platforms leverage AI and machine learning to forecast anomalies, optimize performance, and even self-heal without human intervention. The evolution from reactive operations to predictive intelligence transforms observability into an active control plane that not only detects problems but also orchestrates solutions in real time. But the challenge is ensuring these autonomous actions remain transparent, trustworthy, and aligned with business priorities. Organizations that embrace this proactive model will move from firefighting to foresight, gaining resilience and competitive advantage.

Observability as a business superpower

Observability enables organizations to see across systems, anticipate disruptions, and move from passive monitoring to informed, decisive action before issues impact customers or revenue. When powered by AI and automation, observability becomes a predictive engine for growth and resilience, enabling businesses to innovate faster, operate smarter, and compete with confidence. Observability done right turns complexity into advantage and uncertainty into opportunity.

100% of technology leaders surveyed reported that their organization uses AI as part of it's operations.

THE STATE OF OBSERVABILITY 2025, DYNATRACE

CHALLENGE ONE

Data overload in complex, dynamic AI ecosystems

The pace of technology adoption and the expansion of digital ecosystems have created unprecedented complexity.

Multi-cloud architectures, agents communicating with agents, AI applications, and AI-driven systems generate massive volumes of telemetry at incredible speed. Modern AI stacks introduce orchestration layers, vector databases, and specialized pipelines for model training and inference, each adding new dimensions of complexity. Finally, these systems are not deterministic, making it far harder to predict behavior or troubleshoot issues.

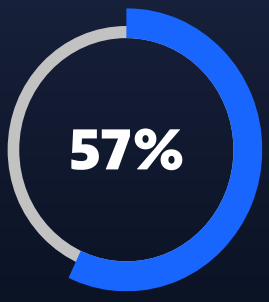


This flood of data makes it nearly impossible for IT teams to understand relationships and dependencies in real time through manual approaches. Without automation and a unified data foundation, observability becomes reactive and incomplete, leaving blind spots that threaten performance and reliability.

Teams often struggle because they are:

- Overwhelmed by the sheer volume and velocity of telemetry.
- Unable to maintain context across dynamic, ephemeral environments.
- Struggling to manage black-box AI systems, where probabilistic behavior obscures how decisions are made and why outputs vary.

These shortcomings introduce risk, slow down digital transformation, and drain resources that could be focused on delivering business value.



of technology leaders reported that their organization is using AI for data management.

THE STATE OF OBSERVABILITY 2025, DYNATRACE



How to overcome it

Achieving observability at scale requires automation, intelligent context, and architectures designed for AI-driven complexity. This means:

- AI-powered analytics on a unified data lakehouse that ingests telemetry in real time and cuts through the noise to surface what matters most.
- A continuously updated context graph with automated discovery and dependency mapping to prevent blind spots.
- An autonomous operations layer grounded in explainable AI that enables trusted decisions and safe automation.

With automation and intelligent context, teams can tame complexity, eliminate data overload, and shift from reactive firefighting to proactive optimization, ultimately accelerating innovation and improving business outcomes.

CHALLENGE TWO

Siloed people and data

Modern enterprises depend on collaboration across infrastructure, development, operations, application, and business teams to deliver digital services at scale. Yet these teams often operate in isolation, with different tools, processes, and priorities.

At the same time, data is fragmented across environments and platforms, creating blind spots and slowing decision-making. The rapid adoption of AI has introduced an explosion of new tools, frameworks, and emerging standards, each adding complexity to already distributed environments. Organizations are ingesting telemetry from countless sources, including applications, cloud services, AI-driven systems, and OpenTelemetry frameworks. But without a connected pipeline, this data remains disjointed and hard to trust.

Teams often struggle because they are:

- Using disparate tools and lacking shared context between technical and business stakeholders.
- Working hard to normalize and enrich data for consistent quality and cost efficiency.
- Unable to rely on AI insights due to fragmented data and a lack of the context AI needs to support accurate decision-making.

These shortcomings lead to inefficiency, duplicated effort, and missed opportunities to optimize performance and deliver value faster.

How to overcome it

Breaking down silos requires a platform approach to observability that unifies data, workflows, and insights across teams. This means:

- Centralizing observability data for a single source of truth and providing role-based views so every stakeholder sees what matters most.
- Building flexible data pipelines that can filter, enrich, and normalize telemetry from sources like OpenTelemetry for better quality and cost control.
- Leveraging AI insights built on a unified data layer and context graph to drive reliable and repeatable actions, while automating collaboration through shared dashboards, alerts, and workflows.

> Observability in the AI era breaks down functional silos and creates a shared intelligence layer across departments for smarter decisions, safer automation, and more efficient operations.
>
> THE STATE OF OBSERVABILITY 2025, DYNATRACE

CHALLENGE THREE

Knowing what efforts drive positive business impact

Digital systems don't just support the business: they ARE the business.

This means that performance issues, inefficiencies, or compliance gaps in technology directly impact revenue, customer satisfaction, and even sustainability goals. Yet most organizations struggle to answer a critical question:

Which efforts actually move the needle for the business?

Observability platforms generate massive volumes of data, but without the ability to correlate technical signals with business outcomes, teams are left guessing. This is especially true with AI systems, where the ROI is often unclear. Are these models improving conversion rates, reducing risk, or simply increasing spend and complexity?

It's no longer enough to monitor uptime or latency. Teams need to understand, in real time, how digital experiences influence revenue, how infrastructure decisions affect margins, and how automation and AI impact compliance, risk, and sustainability. Without this clarity, organizations risk investing in technology that doesn't deliver measurable value.

Teams often struggle because they are:

- Measuring technical metrics without mapping them to KPIs like revenue, cost efficiency, or sustainability.
- Lacking visibility into how IT and AI decisions affect business processes such as FinOps, compliance, and customer experience.
- Unable to unify telemetry with business data and leverage AI-driven insights for real-time impact analysis.

These gaps lead to wasted resources, missed opportunities to innovate, and difficulty justifying technology investments to leadership.



of organizations currently use AI to align observability data with business KPIs, but that number is rising fast.

THE STATE OF OBSERVABILITY 2025, DYNATRACE



How to overcome it

To turn observability into a business superpower, organizations need to:

- Map technical signals to business processes, connecting KPIs such as revenue, cost, compliance and sustainability along with supporting initiatives like FinOps for smarter cloud spend.
- Pinpoint which actions drive outcomes with AI-driven causal analysis and give stakeholders visibility into how technology decisions impact the business.
- Enable real-time business optimization by continuously correlating technical signals with business KPIs and dynamically adjusting decisions to maximize revenue, efficiency, and resilience.

When observability platforms deliver real-time context, troubleshooting shifts from reactive firefighting to strategic decision-making. Instead of just spotting a latency spike, teams can instantly see its projected impact on revenue, customer experience, and even sustainability, turning technical insights into business advantage.

CHALLENGE FOUR

Opacity in AI-driven systems

AI now drives critical business processes, from predictive analytics to autonomous decision-making.

This power introduces complexity and unpredictability, and without transparency, AI's value is hard to prove. When observability platforms lack trustworthy, explainable AI, and when organizations can't monitor their own AI systems effectively, risk increases and trust erodes. Compliance becomes more difficult, costs spiral, and automation can fail in ways that impact customer experience and business outcomes. This challenge becomes even more critical as organizations introduce agentic and multi-agent systems, where autonomous components interact, make decisions, and amplify both impact and risk.

To help AI deliver measurable value, organizations need three things:

1. An observability platform with built-in, explainable, trustworthy AI that can trace decisions, surface causal relationships, and support governance.
2. The ability to monitor AI systems to detect hallucinations, ensure they operate within defined guardrails, and ensure automation aligns with business goals.
3. The ability to track and govern AI agent actions, understand the downstream impact of hallucinations, and trigger machine-speed responses to limit risk and business impact.

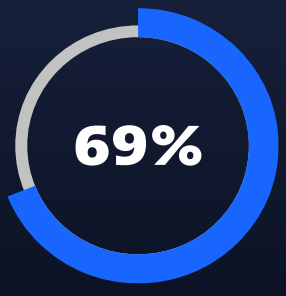
Teams often struggle because they are:

- Blind to how agentic AI layers interact with deterministic systems.
- Unable to validate AI outputs against trusted, unified data foundations.
- Lacking governance to ensure AI actions align with business priorities.
- Finding difficulty proving that automation delivers measurable value.

How to overcome it

- Embed explainability into observability so every AI insight is traceable and auditable.
- Build deterministic foundations using unified, high-quality data with context graphs to understand dependencies and interactions, and to ground AI decisions.
- Balance automation with governance to prevent hallucinations and enforce trust.

When observability makes AI transparent, organizations can innovate confidently, reduce risk, and ensure automation accelerates and doesn't undermine business performance.



of AI-powered decisions are verified by humans

THE PULSE OF AGENTIC AI 2026, DYNATRACE

CHALLENGE FIVE

Automation that lacks context

Automation is essential for scale. To deliver real value, automation must evolve from reactive responses to predictive and proactive actions that anticipate issues and prevent them before they occur. But without context, it becomes a liability. Most systems still react to isolated signals, missing the broader picture. This leads to misfires, cascading failures, and manual intervention that adds complexity instead of reducing it.

Expectations for AI-driven automation are high, but autonomy without accountability introduces risk. Agentic AI doesn't just analyze: it acts. It spins up resources, optimizes workflows, and even makes financial decisions. Without governance and real-time observability, those actions can be misaligned, inaccurate, or costly.

Teams often struggle because they are:

- Automating without visibility into dependencies across applications, cloud services, and business processes.
- Lacking a deterministic foundation to prevent errors and hallucinations, limiting the ability to orchestrate agentic capabilities across platforms and ecosystems.
- Unable to connect automated actions to measurable business outcomes such as cost efficiency, carbon reduction, or customer satisfaction.
- Overloading developers and SRE teams with complexity instead of reducing it.

How to overcome it

- Use observability as the control plane for automation such that every action is informed by real-time context across systems and business processes.
- Ground automation in unified, high-quality data so AI-driven automation is explainable, reliable, and able to safely orchestrate agentic capabilities across platforms and ecosystems.
- Combine predictive intelligence with governance so automation accelerates innovation without compromising compliance or customer experience.
- Design automation to offload complexity from developers and operations teams, freeing them to focus on strategic work.

When automation is powered by observability and context, organizations move from reactive firefighting to predictive and proactive optimization, reducing risk, improving efficiency, and driving measurable business impact.



of technology leaders cite workflow automation as key criterion for selecting an observability platform.

THE STATE OF OBSERVABILITY 2025, DYNATRACE

Dynatrace and Microsoft have forged a deep integration designed to transform complexity from a stumbling block into a strategic asset. By combining Dynatrace AI-powered observability with the Microsoft Azure SRE Agent, organizations can move beyond monitoring to achieve autonomous cloud operations at scale.

This integrated approach enables customers to:

- Contextualize data in real time to uncover precise answers and drive business outcomes faster.
- Boost reliability and resilience with the Azure SRE Agent orchestrating automated remediation informed by Dynatrace causal AI insights.
- Unlock value and accelerate ROI through seamless integration across the Azure ecosystem.
- Empower teams to automate intelligently, reducing manual toil and enabling faster innovation.
- Accelerate AI adoption with advanced observability that ensures transparency, governance, and trust in autonomous actions.

Together, Dynatrace and Microsoft deliver a unified control plane for cloud operations, fueling adoption, consolidating toolsets, and enabling agentic and generative AI to work smarter, safer, and more efficiently.

CONCLUSION

The following foundations of modern observability are necessary to overcome these key challenges:

1. Unified data layer

- A unified data lakehouse with centralized, flexible storage for observability, security, and business signals in real time.
- A single source of truth that supports deterministic analytics and AI on continuously updated data.
- Built-in governance and flexible querying to support retention, investigation, and AI training without data silos.

2. Real time context graph

- A continuously updated context graph that models dependencies across applications, infrastructure, services, and users as the environment changes.
- Causal understanding, not just correlation, explaining what is happening, why it is happening, and where impact will be felt.
- Real-time enrichment and elastic scale to maintain accurate context through ephemeral services, spikes, and high-cardinality telemetry.

3. Autonomous operations system

- Repeatable prevention, remediation, and optimization driven by continuous learning that improves accuracy and reduces noise over time.
- Deterministic and agentic AI working together to deliver explainable answers and take context-aware action, not just generate insights.
- Trusted orchestration of automation and agents across internal platforms and third-party ecosystems using the same integrated data and causal context.



Understand your business like never before

Every business today is a digital business, and success now depends on the ability to simplify real-time decisions by fully leveraging IT and technical data. Unified observability connects technical health, system performance, and overall business outcomes, revealing not only how infrastructure and code behave, but also how those realities impact ROI, customer experience, sustainability, and costs. By unifying these levels of understanding on a single platform, organizations can transform their observability strategy from an operational necessity into a strategic advantage, moving from managing complexity to driving meaningful business impact.

Dig deeper into today's observability ecosystem and learn the five key criteria to consider by downloading the free observability platform buyers' guide.

Download now

Ready to get started now? Explore the Dynatrace Playground for a free, hands-on trial to experience the future of observability today.

ABOUT DYNATRACE

Dynatrace is advancing observability for today's digital businesses, helping to transform the complexity of modern digital ecosystems into powerful business assets. By leveraging AI-powered insights, Dynatrace enables organizations to analyze, automate, and innovate faster to drive their business forward. Learn more at www.dynatrace.com.

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